<u>Proposition</u> For large j $\Phi_{L_Y^j} : Y \longrightarrow W_j \subset \mathbb{CP}^N$ is injective.

<u>Proof</u> We can assume that $\Phi_{L_Y^j}$ have finite fibers

$$\begin{tikzcd}
 & & & Y \arrow[dl] \arrow[dr, "\Phi_{L_Y^{2^j}}"] \arrow[dll, bend right, "\Phi_{L_Y}"'] & & \\
W_1 & W_2 \arrow[l] & W_4 \arrow[l] & \cdots \arrow[l] & W_{2^j} \arrow[l] & \cdots \arrow[l]
\end{tikzcd}$$

all horizontal maps & compositions have finite fibers with a uniform bound

$$\# f_{1,2^j}^{-1}(w) \leq N'$$

$\Rightarrow$ maps eventually stabilize

i.e. eventually $\Phi_{L_Y^{2^j}}$ is injective. $\square$

<u>Further properties</u>

- $$\Phi_{L_Y}(\underset{\substack{\uparrow \\ \text{metric regular set}}}{Y_{reg}}) \subset \underset{\substack{\nwarrow \\ \text{complex analytic regular set}}}{W_{reg}}$$

$\Rightarrow$ $W \setminus W_{reg}$ has codimension ≥ 4 (using e.g. Cheeger-Naber)

Can be used to show that W is normal.

- As consequence, along sequence (M_k, ω_k) maps $\Phi_{L_k^m}$ are embeddings for suitable m & large k

<u>Case of $Ric(\omega_k) \geq -\omega_k$</u> (instead of $|Ric(\omega_k)| \leq 1$)

$(M_k, \omega_k) \xrightarrow{GH} (Y, d_Y)$ non-collapsed, polarized ($-\sqrt{-1}\,\omega_k$ = curvature of $L_k \to M_k$)

- Y can have codim 2 singularities with tangent cone $\mathbb{C}^{n-1} \times C(S^1_\gamma)$

S^1_γ = circle of length $2\pi\gamma$, $\gamma \in (0,1)$

- Singular set $S \subset Y$ can be dense

<u>New ingredient</u>: if $d_{GH}\left(B(p, \varepsilon^{-1}), B(0, \varepsilon^{-1})\right) < \varepsilon$, $p \in M_k$, $0 \in \mathbb{R}^{2n}$

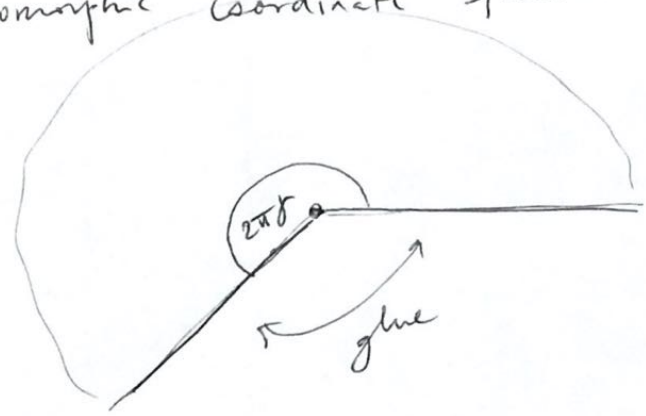
then we have holo. coords $z_1, \dots, z_n$ on $B(p,1)$ and Kähler potential φ for ω_k s.t. $|\varphi - d(p,\cdot)^2| < \Psi(\varepsilon \mid n)$

Using this existence of suitable holo. sections of L_k^m can be shown as before.

- Complex structure on Y is regular around points with tangent cone $\mathbb{C}^{n-1} \times C(S^1_\gamma)$

$\rightsquigarrow$ Y is a normal variety as before

<u>Basic idea</u>: on $C(S^1_\gamma)$ we have a holomorphic coordinate function $W^{\gamma^{-1}}$



<u>Singular Kähler metrics</u>

Let $Y \subset \mathbb{CP}^N$ be a normal variety, s.t.

$$\mathcal{O}_{\mathbb{P}^N}(1)\big|_{Y_{reg}} = K_{Y_{reg}}^{-m} \quad \text{for some } m > 0$$

Let $L_Y = \mathcal{O}(1)|_Y$

Suppose h is a Hermitian metric on L_Y with locally bounded potentials

If locally $h = e^{-m\varphi}$, the curvature $\omega_\varphi = \sqrt{-1}\partial\bar\partial\varphi$ is well defined on Y_{reg}

If ω_φ is positive, it has unique extension to Y. If $\varphi \in L^\infty$, then $\omega_\varphi^n = (\sqrt{-1}\partial\bar\partial\varphi)^n$ is well defined and defines a metric on $K^{-1}_{Y_{reg}}$:

if v = local section of $K^{-1}_{Y_{reg}}$

define $|v|^2 = (v\bar v)\,\omega_\varphi^n$

Curvature of this metric is $Ric(\omega_\varphi)$ (if φ is smooth)

We say that ω_φ is Kähler-Einstein on Y if

$$e^{-\varphi} = \omega_\varphi^n \quad \leftarrow \text{equality of metrics on } K^{-1}_{Y_{reg}}$$

$$\left(\text{or } e^{-m\varphi} = (\omega_\varphi^n)^m \text{ as metrics on } L_Y\right)$$

<u>Note</u>: • If φ is smooth, then taking curvature we have $\omega_\varphi = Ric(\omega_\varphi)$

• It is equivalent to say that ω_φ is a smooth metric on Y_{reg} s.t. $Ric(\omega_\varphi) = \omega_\varphi$ and ω_φ has locally bounded Kähler potentials